

Project Title: Insurance Claims Processing & Settlement System

1. Overview

The **Insurance Claims Processing & Settlement System** enables end-to-end management of insurance claims, including claim registration, document submission, fraud checks, assessment, and settlement disbursement. The system follows an MVC architecture and is compatible with both **Java (Spring MVC)** and **.NET (ASP.NET Core MVC)** frameworks.

The solution contains five main modules:

1. Claim Registration & Policy Validation
2. Document Submission & Verification
3. Claim Assessment & Surveyor Management
4. Fraud Detection & Compliance Checks
5. Claim Settlement & Payment Processing

2. Assumptions

- Deployment is local using **MySQL or SQL Server**.
- Role-based access for **Customer, Claims Officer, Surveyor, Compliance Officer, Admin**.
- ORM tools: **Hibernate** (Java) or **Entity Framework Core** (.NET).
- Fraud rule engine uses a rule-based scoring model.
- Document uploads simulate OCR-based verification.

3. Module-Level Design

3.1 Claim Registration & Policy Validation Module

Purpose: Captures new claim requests and validates policy details.

Controller:

- ClaimController
 - registerClaim()
 - updateClaim()

- getClaimDetails()
- validatePolicy()

Service:

- ClaimService
 - Implements business logic for claim creation and validation.

Model:

- Claim Entity
 - Attributes:
 - claimId (PK)
 - policyId (FK)
 - customerId (FK)
 - claimType
 - claimAmount
 - claimStatus (REGISTERED, UNDER_REVIEW, APPROVED, REJECTED)

3.2 Document Submission & Verification Module

Purpose: Handles submission and verification of claim-related documents.

Controller:

- ClaimDocumentController
 - uploadClaimDocument()
 - validateClaimDocument()
 - getClaimDocuments()

Service:

- ClaimDocumentService
 - Performs document validation and simulated OCR checks.

Model:

- ClaimDocument Entity
 - Attributes:
 - documentId (PK)
 - claimId (FK)

- documentName
- filePath
- verificationStatus (PENDING, VERIFIED, REJECTED)

3.3 Claim Assessment & Surveyor Management Module

Purpose: Supports assignment of surveyors and assessment of claim loss.

Controller:

- AssessmentController
 - assignSurveyor()
 - recordAssessment()
 - getAssessmentDetails()

Service:

- AssessmentService
 - Computes assessment values and manages surveyor tasks.

Model:

- Assessment Entity
 - Attributes:
 - assessmentId (PK)
 - claimId (FK)
 - surveyorId (FK)
 - assessedAmount
 - assessmentRemarks

3.4 Fraud Detection & Compliance Checks Module

Purpose: Performs rule-based fraud detection and compliance verification.

Controller:

- FraudController
 - runFraudCheck()
 - getFraudScore()

Service:

- FraudService

- Applies fraud rules and risk scoring.

Model:

- FraudCheck Entity
 - Attributes:
 - fraudId (PK)
 - claimId (FK)
 - fraudScore
 - riskFlag (LOW, MEDIUM, HIGH)

3.5 Claim Settlement & Payment Processing Module

Purpose: Finalizes claim decisions and executes settlement payouts.

Controller:

- SettlementController
 - approveSettlement()
 - rejectSettlement()
 - processPayment()

Service:

- SettlementService
 - Manages approval workflow and interacts with simulated payment APIs.

Model:

- SettlementLog Entity
 - Attributes:
 - settlementId (PK)
 - claimId (FK)
 - settlementAmount
 - settlementStatus
 - remarks

4. Database Schema

Claim Table

```
CREATE TABLE Claim (  
  claimId INT AUTO_INCREMENT PRIMARY KEY,  
  policyId INT,  
  customerId INT,  
  claimType VARCHAR(50),  
  claimAmount DECIMAL(15,2),  
  claimStatus ENUM('REGISTERED','UNDER_REVIEW','APPROVED','REJECTED'),  
  FOREIGN KEY (policyId) REFERENCES Policy(policyId),  
  FOREIGN KEY (customerId) REFERENCES Customer(customerId)  
);
```

ClaimDocument Table

```
CREATE TABLE ClaimDocument (  
  documentId INT AUTO_INCREMENT PRIMARY KEY,  
  claimId INT,  
  documentName VARCHAR(100),  
  filePath VARCHAR(255),  
  verificationStatus ENUM('PENDING','VERIFIED','REJECTED'),  
  FOREIGN KEY (claimId) REFERENCES Claim(claimId)  
);
```

Assessment Table

```
CREATE TABLE Assessment (  
  assessmentId INT AUTO_INCREMENT PRIMARY KEY,  
  claimId INT,  
  surveyorId INT,  
  assessedAmount DECIMAL(15,2),  
  assessmentRemarks VARCHAR(255),  
  FOREIGN KEY (claimId) REFERENCES Claim(claimId)  
);
```

FraudCheck Table

```
CREATE TABLE FraudCheck (  
  fraudId INT AUTO_INCREMENT PRIMARY KEY,  
  claimId INT,  
  fraudScore INT,  
  riskFlag ENUM('LOW','MEDIUM','HIGH'),  
  FOREIGN KEY (claimId) REFERENCES Claim(claimId)  
);
```

SettlementLog Table

```
CREATE TABLE SettlementLog (  
    settlementId INT AUTO_INCREMENT PRIMARY KEY,  
    claimId INT,  
    settlementAmount DECIMAL(15,2),  
    settlementStatus VARCHAR(50),  
    remarks VARCHAR(255),  
    FOREIGN KEY (claimId) REFERENCES Claim(claimId)  
);
```

5. Local Deployment Details

- Install MySQL/SQL Server, JDK 17/21/24 or .NET SDK 8.0.
- Deployment targets: **Apache Tomcat (Java)** or **Kestrel (.NET)**.

Steps:

1. Clone repository.
2. Execute SQL schema files.
3. Configure application.properties/appsettings.json.
4. Build and launch application.

6. Conclusion

The Insurance Claims Processing & Settlement System provides a complete MVC-based architecture for managing claims efficiently, ensuring document integrity, enforcing compliance, and enabling fast settlement workflows. Local deployment offers a training-friendly environment for developers.